

TADEX ARCHITECTURE & DESIGN SPEC

Exhaustive Web Implementation & UI/UX Design Specification

Complete functional & operational design document for the Tadex automated crypto signal execution platform, user dashboard, provider portal, and admin center.

Author: Principal System Architect & Senior Product Designer

Target Audience: Product Designers, UI/UX Engineers, Frontend Developers

Primary Entry Points Analyzed: FastApi Backend (`bybit_client/billing_api/app.py`), Public API

(`routes/billing.py`), Supabase Schemas (`supabase/migrations/`), Frontend Client (`api-client.ts`)

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1. Master Information Architecture & Workspace Scan

This section outlines the application surface area, user persona roles, state management paradigms, and structural route hierarchy derived from a complete scan of the Tadex repository.

Application Overview & User Roles

CORE ARCHITECTURAL CONTRACT

Tadex is a non-custodial crypto signal execution automation platform tailored for retail traders in Nigeria (NGN), Kenya (KES), and Ghana (GHS). Users connect exchange API keys with **trade-only permissions** (e.g. Bybit). Tadex never takes custody of funds.

User Role Matrix

Role	Target Persona	Access Level & Capabilities
Guest / Unauthenticated	Prospective Trader	Can view landing pages, pricing tiers, provider directory, public terms, and register.
Standard Subscriber	Retail Trader	Can link Bybit API keys, configure risk/leverage, subscribe to signal providers, view live position monitors, and execute automated trades.
Signal Provider	Trader / Community Leader	Can publish trading signals manually or via Telegram/AI parser, set up subscription plans, monitor subscriber counts, and view revenue analytics.
Platform Administrator	Tadex Ops Team	Has full system access: user management, provider moderation, manual crypto payment verification, system kill-switch, audit logs, and performance metrics.

Master Route Architecture

FRONTEND NAVIGATION TREE

Route Path	Screen Name	Access Tier	Core Functionality
/login	Authentication / Sign In	Public	User login, OTP/Password verification, JWT issue
/register	Registration & Onboarding	Public	Account creation, country selection (NGN, KES, GHS), Terms agreement
/dashboard	Trading Control Center	Authenticated User	Real-time account status, connected keys, position cards, kill-switch button
/exchanges	Exchange Key Management	Authenticated User	Bybit API key input, secret encryption, permission check, environment toggle
/providers	Provider Marketplace	Authenticated User	Provider list, strategy stats, win rate, subscription drawer & checkout
/billing	Billing & Subscriptions	Authenticated User	Active plan details, multi-currency checkout modal, payment history
/risk-settings	Risk Management Center	Authenticated User	Margin sizing per trade, max daily loss limit, default TP/SL, leverage lock
/positions	Positions & Trade History	Authenticated User	Live open positions, PnL indicators, emergency close position modal
/provider-center	Signal Creator Portal	Signal Provider	Channel linking, AI signal builder, subscriber analytics, plan builder
/admin/dashboard	Admin Control Center	Platform Admin	System heartbeat, total volume, subscriber count, emergency global kill switch
/admin/users	User Management Console	Platform Admin	User directory, filter by active/suspended/deleted, view details drawer, suspend action
/admin/providers	Provider Moderation Console	Platform Admin	Provider applications, verification badges, plan approval, suspend/delete provider
/admin/billing	Billing & Crypto Review	Platform Admin	Manual crypto/USDT payment verification modal, plan grant overrides, invoice audit

2. Global Layout & Navigation Specifications

Global Sticky Top Header

LAYOUT COMPONENT

Visual Structure & Grid

Sticky 64px top bar anchored across the full viewport. Contains brand logo on the left, environment indicator (Demo vs Live), real-time status pill, regional currency selector, notification drawer bell, and user avatar dropdown on the right.

Granular UI Elements

- **Brand Logo & Tagline:** Tadex logo with a subtle glowing status dot. Clicking routes to `/dashboard`.
- **Environment Badge Pill:** Interactive pill displaying "LIVE MAINNET" (emerald theme) or "TESTNET / DEMO" (amber theme). Clicking opens a confirmation modal to switch trading environments safely.
- **Engine Real-time Heartbeat Indicator:** Blinking green dot with text "Engine Connected". If WebSocket/API connection drops, converts to blinking red with text "Reconnecting...".
- **Currency Selector Pill:** Displays user currency symbol (e.g. 🇳🇮 NGN, 🇰🇪 KES, 🇬🇭 GHS, 🇺🇸 USDT). Hovering displays current exchange rate tooltip against USD.
- **Notifications Bell:** Displays red badge count for unread trade execution alerts or billing warnings. Clicking toggles the slide-out notifications panel.
- **User Avatar Menu:** Displays user avatar or initials. Click dropdown reveals: Profile Settings, Subscription Status ("Pro Plan - 14 days left"), Dark/Light Mode toggle, and Sign Out.

DESIGNER NOTE: REAL-TIME HEARTBEAT & TOKEN REFRESH

The UI uses in-memory JWT tokens and background WebSockets. The heartbeat indicator lets the user know trade execution is active without needing to refresh. If the session expires (HTTP 401), the app performs a silent background refresh using an `httpOnly` refresh cookie (`api-client.ts`). If refresh fails, a subtle full-screen session expired modal appears.

Collapsible Navigation Sidebar

[LAYOUT COMPONENT](#)

Visual Hierarchy

Vertical sidebar (240px expanded, 64px collapsed) pinned to the left. Grouped into three distinct semantic navigation blocks: **Trading**, **Provider Center**, and **System Administration** (visible only to admins).

Navigation Items

- **Dashboard:** Home icon. High-level metric overview and active kill switch.
- **Exchange Keys:** Key icon. Status badge showing connected exchange status (e.g., "Bybit Connected ").
- **Marketplace:** Compass icon. Explore top signal providers with live win-rate tags.
- **Positions:** Layers icon. Shows live badge count of open trades.
- **Risk Controls:** Shield icon. Sizing rules, daily stop limits, and leverage defaults.
- **Billing & Plans:** Credit Card icon. Current plan expiry and invoice history.
- **Admin Console (Admin Only):** Lock-Shield icon with purple highlight. Opens user and provider management sub-items.

3. Core Application Pages Specification

Screen 3.1: User Trading Dashboard

CORE APP SCREEN

1. Screen Purpose & Context

The primary control center for retail traders. Provides instant visibility into account balance, active position health, automated signal execution logs, and emergency risk controls.

2. Layout & Page Structure

Top section: 4 Stat Cards grid. Middle left (65% width): Active Positions & Execution Stream. Middle right (35% width): Quick Risk Controls & Active Subscriptions. Bottom: Emergency Kill Switch Banner.

3. Granular UI Elements & Dynamic States

- **Total Portfolio Value Card:** Displays account balance in USD and user currency (e.g. `\$2,450.00 ≈ ¥3,920,000`). Hovering displays free margin vs locked margin breakdown.
- **Daily PnL Card:** Displays percentage change with color coding (Green for +%, Red for -%). Includes a micro sparkline chart showing today's equity curve.
- **Connected Exchange Card:** Shows Bybit logo, API key label ("Bybit Main Trades"), and status indicator ("Connected & Validated"). If key invalid, shows warning badge with "Reconnect Required" button.
- **Global Emergency Kill Switch Button:** Prominent crimson button with shield icon: *"PAUSE ALL AUTOMATED TRADING"*. Clicking opens confirmation modal: "This will pause signal execution immediately. Existing open positions can be kept or closed."
- **Live Open Positions Table:** Displays Symbol (e.g. BTCUSDT), Side (LONG/SHORT badge), Entry Price, Current Price, Sizing, and Unrealized PnL.
 - *Empty State:* "No open positions. Signals from your subscribed providers will execute automatically."
 - *Hover State:* Highlight row with quick actions: "Close Position" button.

DESIGNER NOTE: POSITION MONITORING & IDEMPOTENCY

Trade executions are driven by backend background tasks. When a signal arrives, the engine performs preflight checks (risk size, leverage limits, balance checks). Positions update via live stream (`monitoring/position_monitor_reconciliation.py`). The UI must show a subtle pulse effect on position rows when updated.

Screen 3.2: Exchange Key Management

CORE SECURITY SCREEN

1. Screen Purpose & Context

Allows users to securely connect their exchange API credentials (e.g., Bybit). Enforces strict security guidance: API keys must have **trade-only** permissions, disabling withdrawal capabilities.

2. Layout & Page Structure

Two-column layout. Left column (60%): Exchange API Key Input Form. Right column (40%): Security Guidelines & Step-by-Step Tutorial Box.

3. Granular UI Elements & Dynamic States

- **Exchange Selection Cards:** Selectable radio cards for exchanges. Bybit highlighted as primary supported exchange.
- **API Key Input Field:** Text input with copy-paste helper. Masked for security.
- **API Secret Input Field:** Password field with eye toggle to view/hide. Never saved in plaintext.
- **Account Environment Radio:** Toggle between "Live Mainnet" and "Testnet/Demo Account".
- **Validate & Save Button:** Primary action button. Clicking triggers a real-time API handshake with Bybit (`bybit_client/executor/exchange_validator.py`) to verify key permissions.
 - *Loading State:* Spinner icon with text "Validating Bybit API Permissions...".
 - *Success State:* Green toast banner: "Bybit API Key verified! Trade-only permissions confirmed."
 - *Error State (Withdrawal Enabled Warning):* Red warning alert box: "**Security Violation: Withdrawal permission detected on key. Tadex requires trade-only keys. Please edit your key on Bybit.**"

Screen 3.3: Provider Marketplace & Checkout

CORE MARKETPLACE SCREEN

1. Screen Purpose & Context

Marketplace where traders discover, evaluate, and subscribe to signal providers. Provides performance metrics, win rates, historical drawdowns, and flexible multi-currency pricing plans.

2. Layout & Page Structure

Top bar: Search input, Filter by Win Rate, Filter by Market (Spot/Futures), and Sort by Popularity. Main area: Responsive 3-column card grid. Clicking a provider card opens the Provider Detail & Subscription Drawer.

3. Granular UI Elements & Components

- **Provider Card Header:** Avatar, Provider Name, Verification Badge ("Verified Provider )", and Telegram Channel Link tag.
- **Performance Stats Grid:**
 - Win Rate (e.g. 78.4%)
 - 30-Day Total Signals (e.g. 42 signals)
 - Max Drawdown (e.g. -6.2%)
 - Active Subscribers Count (e.g. 340 traders)
- **Subscribe Button:** Primary button displaying monthly price in user's currency (e.g. "Subscribe from ₦15,000/mo").

4. Multi-Currency Checkout Drawer Modal

When user clicks "Subscribe", a side drawer slides out from the right:

- **Duration Selector Buttons:** 1 Month, 3 Months (5% off), 6 Months (10% off), 12 Months (20% off).
- **Platform Tier Selection:** Basic Platform Tier vs Pro Execution Tier (`platform_tiers` table).
- **Currency Settlement Selector:** Toggle between NGN (Flutterwave/Paystack), KES (M-Pesa/Flutterwave), GHS (Mobile Money), or USDT (Direct Crypto Transfer).
- **Price Breakout Summary:** Shows Provider Plan Monthly Fee + Platform Monthly Fee = Final Total Amount in selected currency.
- **Proceed to Payment Button:** Initializes checkout (`POST /billing/init`). Redirects to gateway or opens manual crypto deposit modal with QR code and transaction hash input.

DESIGNER NOTE: MULTI-CURRENCY CHECKOUT LOGIC

The backend API (`billing_api/routes/billing.py`) calculates pricing based on provider plan duration and platform tier quotes. For crypto manual payments, it displays an exact USDT wallet address and reference code. Upon user submission, the status transitions to "Pending Admin Verification".

4. Signal Provider Creator Portal Specification

Screen 4.1: AI Signal Builder & Parser

PROVIDER PORTAL SCREEN

1. Screen Purpose & Context

Empowers signal providers to type or paste signal text in natural language (e.g. from Telegram) and automatically extract symbol, direction, entry targets, stop loss, and take profit targets using Tadex's AI signal parser.

2. Layout & Page Structure

Left pane (50%): Raw Text Input Editor. Right pane (50%): Extracted Signal Struct Card & Execution Preview.

3. Granular UI Elements & Workflow

- **Raw Signal Textarea:** Input box with placeholder text: *"Paste signal e.g. BUY BTCUSDT ENTRY 64000 TP1 65500 SL 62500..."*
- **AI Extract Button:** Triggers backend signal extraction (`ai_signal_builder.py`).
- **Extracted Candidate Card:** Form fields pre-filled by parser:
 - Symbol Dropdown (e.g., BTCUSDT)
 - Trade Direction Badge (LONG / SHORT)
 - Entry Price Range (Single price or multi-target entry)
 - Take Profit Targets (TP1, TP2, TP3 addable rows)
 - Stop Loss Input with Risk/Reward Ratio calculator badge (e.g. "R:R = 1:2.4")
- **Broadcast Signal Button:** Green primary button: *"Confirm & Broadcast Signal to Subscribers"*. Shows confirmation alert before broadcasting to all subscriber execution queues.

5. Admin Portal Specification

The administrative portal provides operational tools for platform managers to monitor system health, manage users, moderate providers, review crypto payments, and enforce global safety controls.

Screen 5.1: Admin Master Dashboard

ADMIN PORTAL

1. Screen Purpose & Context

Operational control center for Tadex system admins. Provides real-time metrics on total volume executed, active subscribers, system error rates, and system-wide kill switch overrides.

2. Layout & Page Structure

Top section: 4 Key Performance Metric Cards. Middle: Execution Traffic Chart & Error Rate Stream. Right: System Health Checks & Quick Action Panel.

3. Granular UI Elements & Controls

- **Metric Cards:**
 - Total Registered Users (`admin_users` count)
 - Active Paid Subscriptions (Active/Trialing status)
 - 24-Hour Trading Volume (\$ USD equivalent)
 - Pending Payment Approvals (**REQUIRES ACTION**)
- **Master System Kill Switch Button:** Large red action toggle: **"SYSTEM-WIDE TRADING PAUSE"**. Requires dual confirmation modal with admin password re-entry to halt all worker execution threads (`bybit_client/executor/worker_pool.py`).
- **System Log Severity Counter:** Real-time counter of INFO, WARNING, ERROR, and CRITICAL logs (`test_events_log_severity_normalization.py`).

Screen 5.2: Admin User Management Console

ADMIN PORTAL

1. Screen Purpose & Context

Allows admins to search, filter, inspect, suspend, unsuspend, or soft-delete user accounts (`bybit_client/telegram/handlers/admin_user.py`).

2. Layout & Page Structure

Top filter toolbar: Search bar (Telegram ID, Username, Email), Status Filter Tabs (All,  Active,  Suspended,  Deleted). Main table: Paginated User List. Right: Slide-out User Detail Drawer.

3. Granular UI Elements & Table Columns

User Identifier	Status	Created Date	Connected Exchanges	Active Subscriptions	Actions
@trader_ng TG: 58291039	 ACTIVE	2026-04-12	1 (Bybit)	2 Providers	<button>View Details</button> <button>Suspend</button>
@crypto_master TG: 19482014	 SUSPENDED	2026-02-18	0 Keys	0 Plans	<button>View Details</button> <button>Unsuspend</button>

4. User Action Modals & Drawers

- User Detail Drawer:** Displays complete user telemetry: trade history count, last active timestamp, restriction reasons, and assigned platform tier.
- Suspend User Modal:** Input field for mandatory restriction reason: *"Reason for suspension..."*. Action buttons: "Confirm Suspension" (triggers `suspend_user()`) and "Cancel".
- Soft Delete Modal:** Warning dialog: "Soft-deleting this account will hide it from operations while preserving all historical trade logs for compliance."

Screen 5.3: Admin Provider Moderation Console

[ADMIN PORTAL](#)

1. Screen Purpose & Context

Interface for admins to review provider applications, verify provider credentials, set custom fee shares, and manage provider status (`bybit_client/telegram/handlers/admin_provider.py`).

2. Granular UI Elements

- **Pending Applications Queue:** List of new signal providers awaiting review. Shows trading experience, Telegram channel link, and proof of performance.
- **Approve / Reject Action Buttons:** "Approve Provider" assigns provider role and enables plan creation. "Reject Application" sends notification with rejection feedback.
- **Verification Toggle:** Admin toggle to assign or revoke the "Official Verified Provider" badge.
- **Provider Status Actions:** "Suspend Provider" (pauses provider signal distribution), "Restore Provider", and "Soft-Delete Provider".

Screen 5.4: Admin Manual Crypto Payment Review

[ADMIN PORTAL](#)

1. Screen Purpose & Context

Provides administrative oversight for manual crypto/USDT deposits. Admins review submitted transaction hashes against blockchain receipts and approve or reject subscription activations (`admin_billing.py`).

2. Granular UI Elements

- **Pending Crypto Submissions Table:** Columns: User ID, Provider Plan, Amount (USDT), TX Hash, Submission Time, Status ("Pending Review").
- **Blockchain Verification Link:** External link opening Etherscan/Tronscan for the submitted TX Hash.
- **Approve Payment Button:** Triggers backend subscription activation (`admin_crypto_approve`), updating subscription status to "Active" and generating a receipt.
- **Reject Payment Modal:** Rejection reason options: "Transaction hash not found", "Insufficient amount", or "Duplicate hash". Notifies user via app alert and Telegram.

6. Edge Cases & System States Specification

System Error & Edge Case Modals

SAFETY & ERROR INTELLIGENCE

1. Session Expiration Overlay

When access token refresh fails, a full-screen blurred backdrop modal blocks UI interactions: **"Session Expired. Please sign in again to continue trading safely."** Includes a single "Sign In" button routing to `/login`.

2. Exchange Leverage Cap Warning

If a provider signal specifies leverage exceeding user's custom risk setting or Bybit account tier, the execution engine triggers a leverage override (`utils/leverage_protector.py`). The UI displays an warning toast: **"Leverage Capped: Signal requested 50x, executed at safe max 20x."**

3. Insufficient Balance Preflight Modal

When an automated trade fails due to low exchange balance, an inline warning card appears on the dashboard: **"Execution Skipped: Insufficient USDT Margin on Bybit. Deposit funds or adjust position sizing in Risk Settings."**

4. Loading Skeletons & Empty States

- **Table Loading State:** Animated shimmer cards matching exact row height to eliminate cumulative layout shift (CLS).
- **Empty Data States:** Illustration icon with actionable text (e.g. "No connected exchanges yet — [Connect Bybit Key]").